

❄️ Snowflake Learning Series – SnowSQL (Real-Time End-to-End Project) ❄️

Now that we understand **SnowSQL installation, configuration, commands, and variables**, let's see how it is used in a **real production project**.

This example shows how **Data Engineers** automate data ingestion into **Snowflake** using **SnowSQL**.

Real-Time Scenario

Assume you are working in an **E-commerce company**.

Every day the company receives **customer order data as CSV files** from different systems.

Example file received daily:

orders_2026_03_10.csv

This file must be loaded into **Snowflake Data Warehouse** for:

- Sales analytics
- Customer insights
- Business dashboards
- Reporting tools like Tableau / Power BI

Instead of manually loading the file every day, we **automate the entire pipeline using SnowSQL**.

End-to-End Architecture

Source System

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CSV Files (Daily Data)

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SnowSQL Script

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Snowflake Stage

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Snowflake Table

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BI / Analytics Tools

Step 1 – Connect to Snowflake using SnowSQL

First we connect to Snowflake from the terminal.

```
snowsql -a company123.ap-south-1 -u data_engineer
```

Explanation:

- **-a** → Snowflake account name
- **-u** → username

Once executed, SnowSQL will open a session.

Example prompt:

```
data_engineer#COMPUTE_WH@DEMO_DB.PUBLIC>
```

This means we are now connected to Snowflake.

Step 2 – Select Warehouse, Database, and Schema

Before loading data we must define the working environment.

```
USE WAREHOUSE COMPUTE_WH;  
USE DATABASE SALES_DB;  
USE SCHEMA PUBLIC;
```

Explanation:

Warehouse

Compute engine used to run queries.

Database

Logical container that stores data.

Schema

Organizes tables inside the database.

Step 3 – Create Table to Store Data

Now create a table where the CSV file data will be loaded.

Example: Orders table.

```
CREATE OR REPLACE TABLE ORDERS (  
  ORDER_ID STRING,  
  CUSTOMER_ID STRING,  
  PRODUCT_ID STRING,  
  ORDER_DATE DATE,  
  QUANTITY NUMBER,  
  PRICE NUMBER  
);
```

Explanation:

Each column represents fields from the CSV file.

Example CSV data:

101,CUST01,P100,2026-03-10,2,500

102,CUST02,P200,2026-03-10,1,1200

This data will be inserted into the ORDERS table.

Step 4 – Upload File to Snowflake Stage

Snowflake uses **staging areas** to store files temporarily before loading.

We upload the file using the **PUT command** in SnowSQL.

```
PUT file:///data/orders_2026_03_10.csv
@%ORDERS;
```

Explanation:

- **PUT** → Uploads file from local machine
- **@%ORDERS** → Table stage linked to ORDERS table

So the CSV file is now stored in Snowflake stage.

Step 5 – Load Data into Table

Now we load the staged file into the table using **COPY INTO command**.

```
COPY INTO ORDERS
FROM @%ORDERS
FILE_FORMAT = (
  TYPE = CSV
  FIELD_OPTIONALLY_ENCLOSED_BY = ''
  SKIP_HEADER = 1
);
```

Explanation:

COPY INTO

Bulk loading command in Snowflake.

TYPE = CSV

File format type.

SKIP_HEADER = 1

Ignores the first row (column names).

Step 6 – Validate Loaded Data

Once the data is loaded we verify the records.

```
SELECT * FROM ORDERS;
```

Or count rows.

```
SELECT COUNT(*) FROM ORDERS;
```

Example output:

3,500 rows loaded successfully

Now the data is available for analytics.

Step 7 – Automating the Pipeline using SQL Script

Instead of running commands manually, Data Engineers create a script.

Example:

load_orders.sql

Script contents:

```
USE WAREHOUSE COMPUTE_WH;  
USE DATABASE SALES_DB;  
USE SCHEMA PUBLIC;
```

```
PUT file:///data/orders_2026_03_10.csv  
@%ORDERS;
```

```
COPY INTO ORDERS  
FROM @%ORDERS  
FILE_FORMAT = (  
  TYPE = CSV  
  SKIP_HEADER = 1  
);
```

Run the script using SnowSQL.

```
snowsql -f load_orders.sql
```

Explanation:

- -f executes SQL file
 - Entire data pipeline runs automatically.
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Step 8 – Scheduling the Pipeline

In real projects this SnowSQL script is scheduled using:

- **Airflow**
- **Cron Jobs**
- **Control-M**
- **Azure Data Factory**
- **Informatica**

Example workflow:

Daily 2 AM

New CSV file arrives

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Scheduler triggers SnowSQL script

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File uploaded to Snowflake

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COPY INTO loads data

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Dashboards updated automatically

Final Data Flow

CSV File

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SnowSQL PUT

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Snowflake Stage

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COPY INTO

↓

Snowflake Table

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BI Dashboard

Why Companies Use SnowSQL in Production

- ✓ Automates data ingestion
 - ✓ Easy scripting support
 - ✓ Works across Windows/Linux/Mac
 - ✓ Integrates with schedulers
 - ✓ Supports large file loading
 - ✓ Ideal for data pipeline automation
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Key Learning Summary

SnowSQL is widely used by **Data Engineers** to:

- Connect Snowflake via CLI
- Execute SQL queries
- Automate ETL pipelines
- Upload files using PUT
- Load data using COPY INTO
- Schedule scripts for daily processing

This makes SnowSQL a **powerful automation tool in the Snowflake ecosystem.**